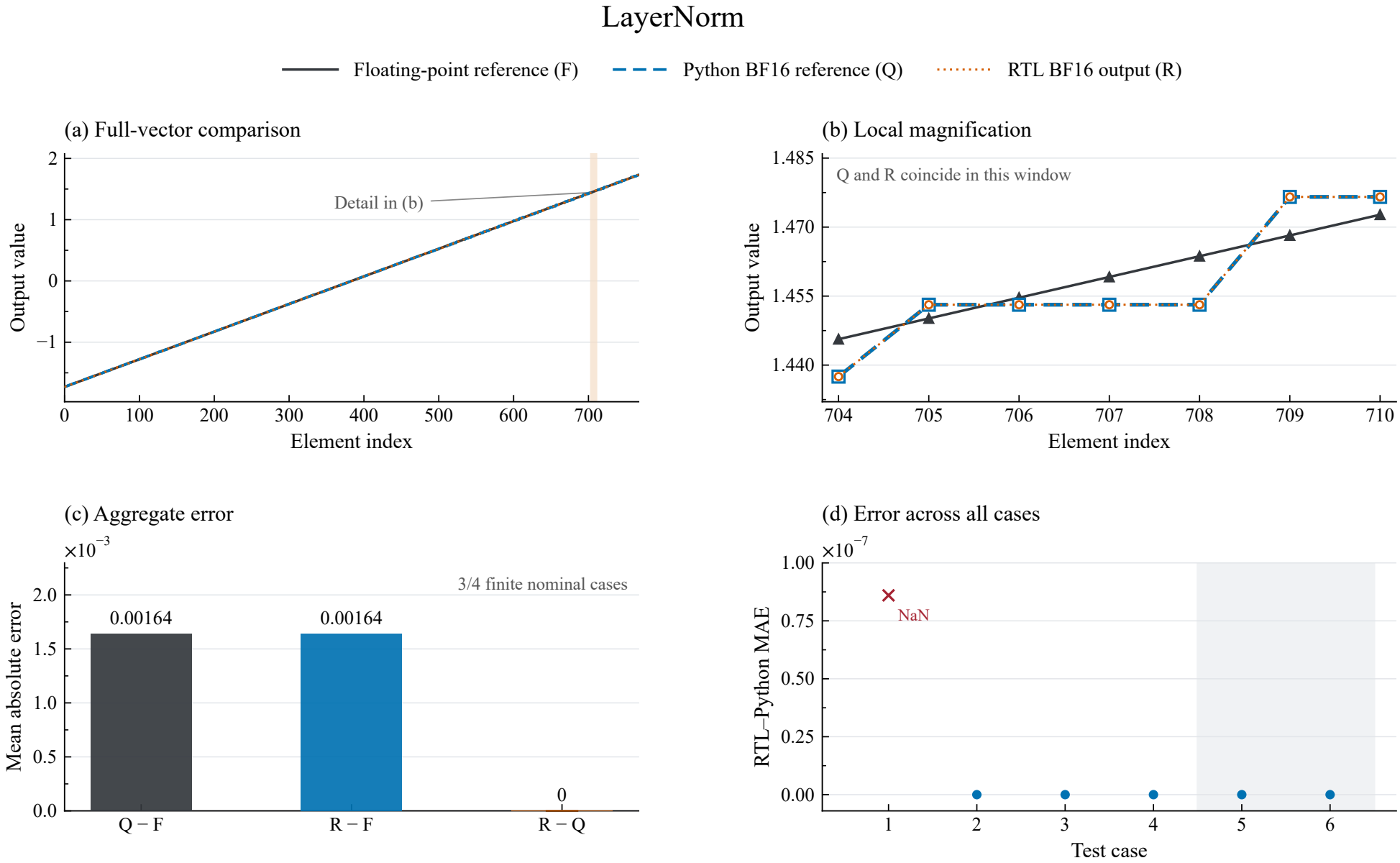
LayerNorm
Floating-point reference (F)
Python BF16 reference (Q)
RTL BF16 output (R)
(a) Full-vector comparison
Output value
Element index
Detail in (b)
(b) Local magnification
Q and R coincide in this window
Output value
Element index
(c) Aggregate error
Mean absolute error
3/4 finite nominal cases
0.00164
0.00164
0
Q − F
R − F
R − Q
(d) Error across all cases
RTL–Python MAE
NaN
Test case
GEMM + operator · N = 768 · ramp example #3 · shaded cases: probes · red markers: failed checks
All-case RTL MAE is undefined (1 nonfinite nominal case).